Jonas H. Møller (CV)

Applied Mathematics Student

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Profile

Applied Mathematics student at DTU interested in machine learning, statistical modeling, and data-driven systems. Experience building and implementing mathematical models through academic projects involving stochastic processes, optimization, and machine learning. I enjoy learning about and combining mathematics and programming to build and evaluate models for complex data.

Education

Technical University of Denmark (DTU) *2023 – Present*
Bachelor of Science in Engineering (Applied mathematics)
Selected courses: Mathematical Modeling (12), Academic Project (Fagprojekt) (12), Statistical Modeling (12), Discrete Mathematics 2 (12), Optimization and Data Fitting (10), Introduction to Machine Learning and Data Mining (10), Introduction to reinforcement learning and control theory (10)
Note: number in parentheses denotes achieved grade in the Danish 7-point grading scale.

University of Copenhagen (KU) *2022 – 2023*
Coursework in Mathematics and Economics

TEC — HC Ørsted Gymnasiet (HTX) *2018 – 2021*
Technical Upper Secondary Education — Mathematics and Physics (Chemistry A)

Technical Skills

Programming: Python, R, C, F#
Mathematics: Linear algebra, probability, statistics, optimization, numerical methods
Data & ML: Statistical modeling, machine learning, data analysis, visualization (matplotlib, R)
Other: Basic Git, LaTeX, Rcpp

Academic Projects

Bachelor's Thesis (2026): A stability-based investigation of K-means with potential application to disease trajectories *(Grade: pending)*

- Designed simulation experiments to analyze clustering stability under varying data-generating conditions.
- Investigated how cluster separation, symmetry, dimensionality and noise influence clustering robustness, stability and reproducibility.
- Applied the developed methodology to a real-world multimorbidity dataset.

Academic Project (Fagprojekt): Modeling Railway Track Defects (DTU, group project) *(Grade: 12)*

- Implemented a data-driven stochastic model for railway defect progression using continuous-time Markov jump processes
- Estimated transition dynamics from interval-censored data using maximum likelihood and Monte Carlo EM
- Incorporated covariates via Cox regression and performed model selection using AIC

– Implemented computationally efficient estimation routines in R and Rcpp for stochastic process models

– Evaluated forecasting performance through simulation studies.

Machine Learning Projects (DTU)

– Applied supervised learning methods in Python, including logistic regression with L2 regularization.

– Built a classifier for pneumonia detection from chest X-ray images using handcrafted image features based on gradient orientations

– Performed feature extraction, model training, and evaluation using scikit-learn and Python data analysis tools

– Applied clustering and other unsupervised learning methods as part of coursework.

Experience

McDonald's — Crew Member *2021 – 2022*

Worked in a fast-paced team environment, developing reliability, responsibility, and collaboration skills.